# Rayan Alam

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## EDUCATION

**Master of Science in Mechanical Engineering** *Expected Dec 2026*
**University of California, Los Angeles**

**Bachelor of Science in Biomedical Engineering – *Cum Laude*** **May 2025**
**Arizona State University**

Dean's List: Fall 2021- Fall 2022, Spring 2024-Spring 2025

## SKILLS

- **CAD & Simulation:** SolidWorks, Fusion 360, AutoCAD
- **Analysis & Imaging:** Fiji, flow simulation tools
- **Prototyping:** 3D printing, silicone molding, bench testing
- **Software & Tools:** MATLAB, Microsoft Office, Adobe Illustrator, Photoshop

## PROJECTS & RESEARCH

**Bionics Lab, UCLA- Research Assistant** **September 2025 - Present**

- Participate in research on medical and wearable robotics, with emphasis on biomechanics, control systems, and human–machine interaction.
- Contribute to projects involving control systems, haptics, virtual reality, and teleoperation for rehabilitation applications.
- Assist in experimental setup, data collection, and system integration for human-subject studies.

**Implantable Leptin Delivery Pump-Prototype Lead** **August 2024 - May 2025**

- Led development of an implantable pump system designed to deliver leptin into cerebrospinal fluid to prevent Type II diabetes.
- Designed a full parametric CAD assembly in SolidWorks, including a custom silicone bellow to enable controlled, low-flow drug delivery.
- Engineered system to deliver 0.3 mL/day over 6 months, maintaining dosing accuracy within ±0.1 mL of target volume.
- Built and tested prototypes using 3D-printed components and molded silicone, performing leak testing and failure analysis to guide design iterations.

**Weaver Lab: Biomaterials and Immune Engineering-Research Assistant** **August 2023 - May 2025**

- Supported development of islet macroencapsulation devices to improve safety and usability of cell-based therapies for Type 1 diabetes.
- Performed flow simulations in SolidWorks to evaluate mass transport and inform device geometry optimization.
- Tested device effectiveness using Alignate and gas diffusion layer (GDL) materials, exploring design improvements through comparative evaluation.

**CleanWater Toilets-Prototype Designer** **August 2021 -December 2021**

- Designed low-cost sanitation systems aimed at reducing open defecation in rural India, improving hygiene and public health outcomes.
- Created detailed Fusion 360 CAD assemblies to simplify manufacturing and assembly for non-technical stakeholders.
- Developed and pitched a full business plan, including market research, user needs assessment, and scalability projections.

## EXPERIENCE

**Massachusetts General Hospital (MGH) - Harvard Medical School-Research Assistant** **June 2024 - August 2024**

- Supported translational research on enteric neural stem cell (ENSC) transplantation in a mouse model of Hirschsprung disease.
- Assisted in isolating, expanding, and microinjecting ENSCs into Ednrb knockout mice.
- Conducted literature reviews and contributed to histological analysis, observing reduced inflammation and improved survival outcomes in ENSC-transplanted mice.